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Studierichting: Informatica
Oefeningenreeks 5K nr. 1.4

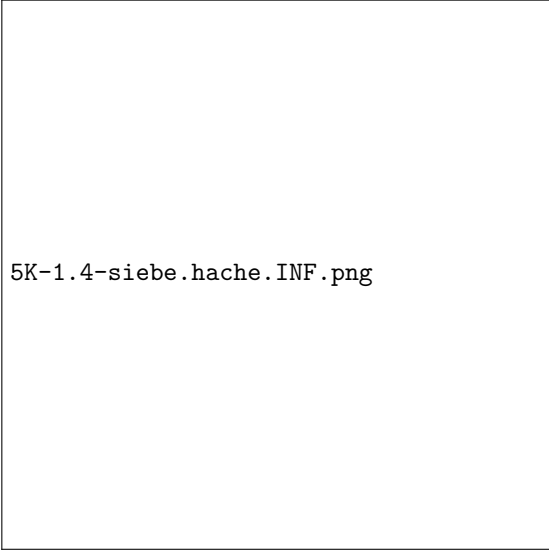
Bereken de booglengte L van de kromme $r = \sin^2\left(\frac{\theta}{2}\right)$ op het interval $[0, 2\pi]$.

afgeleide:

$$\begin{aligned}r' &= 2 \sin\left(\frac{\theta}{2}\right) \cdot \cos\left(\frac{\theta}{2}\right) \cdot \frac{1}{2} = \sin\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right) \\r^2 + (r')^2 &= \left(\sin^2\left(\frac{\theta}{2}\right)\right)^2 + \left(\sin\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right)\right)^2 \\&= \sin^4\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right) \cos^2\left(\frac{\theta}{2}\right) \\&= \sin^2\left(\frac{\theta}{2}\right) \left(\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)\right) \\&= \sin^2\left(\frac{\theta}{2}\right) \cdot 1 = \sin^2\left(\frac{\theta}{2}\right)\end{aligned}$$

Integraal:

$$\begin{aligned}L &= \int_0^{2\pi} \sin\left(\frac{\theta}{2}\right) d\theta \\&= \left[-2 \cos\left(\frac{\theta}{2}\right)\right]_0^{2\pi} \\&= \left(-2 \cos\left(\frac{2\pi}{2}\right)\right) - \left(-2 \cos\left(\frac{0}{2}\right)\right) \\&= (-2 \cos \pi) - (-2 \cos 0) \\&= (-2 \cdot (-1)) - (-2 \cdot 1) = 2 + 2 = 4\end{aligned}$$



5K-1.4-siebe.hache.INF.png

References

- [1] Grafiek gemaakt met geogebra